

6.3.1 Initial insights ¶

6.3.1 Backward error of dot product, simple example

fit width



Before giving a general result, let us focus on the case where the vectors x and y have only a few elements:

Example 6.3.1.1. Consider

$$x = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} \text{ and } y = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}$$

and the computation

$$\kappa := x^T y.$$

Under the SCM given in [Subsubsection 6.2.3.2](#), the computed result, $\tilde{\kappa}$, satisfies

$$\tilde{\kappa} = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) & 0 \\ 0 & (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \end{pmatrix} \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}. \tag{6.3.1}$$

▼ [Solution](#)

$$\begin{aligned} \tilde{\kappa} &= \langle \tilde{\kappa} = [x^T y] \rangle \\ &= \left[\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \right] \\ &= \langle \text{definition of } x^T y \rangle \\ &= [\chi_0 \psi_0 + \chi_1 \psi_1] \\ &= \langle \text{each suboperation is performed in floating point arithmetic} \rangle \\ &= [[\chi_0 \psi_0] + [\chi_1 \psi_1]] \\ &= \langle \text{apply SCM multiple times} \rangle \\ &= \left[\chi_0 \psi_0 (1 + \epsilon_*^{(0)}) + \chi_1 \psi_1 (1 + \epsilon_*^{(1)}) \right] \\ &= \langle \text{apply SCM} \rangle \\ &= (\chi_0 \psi_0 (1 + \epsilon_*^{(0)}) + \chi_1 \psi_1 (1 + \epsilon_*^{(1)}))(1 + \epsilon_+^{(1)}) \\ &= \langle \text{distribute} \rangle \\ &= \chi_0 \psi_0 (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) + \chi_1 \psi_1 (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \\ &= \langle \text{commute} \rangle \\ &= \chi_0 (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) \psi_0 + \chi_1 (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \psi_1 \\ &= \langle \text{(perhaps too) slick way of expressing the final result} \rangle \\ &= \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) & 0 \\ 0 & (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \end{pmatrix} \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \\ &= \end{aligned}$$

where $|\epsilon_*^{(0)}|, |\epsilon_*^{(1)}|, |\epsilon_+^{(1)}| \leq \epsilon_{\text{mach}}$.

An important insight from this example is that the result in [\(6.3.1\)](#) can be manipulated to associate the accumulated error with vector x as in

$$\tilde{\kappa} = \begin{pmatrix} \chi_0 (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) \\ \chi_1 (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}$$

or with vector y

$$\tilde{\kappa} = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} \psi_0 (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) \\ \psi_1 (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \end{pmatrix}.$$

This will play a role when we later analyze algorithms that use the dot product.

Homework 6.3.1.1. Consider

$$x = \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix} \text{ and } y = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{pmatrix}$$

and the computation

$$\kappa := x^T y$$

computed in the order indicated by

$$\kappa := (\chi_0\psi_0 + \chi_1\psi_1) + \chi_2\psi_2.$$

Employ the SCM given in Subsubsection 6.2.3.2, to derive a result similar to that given in (6.3.1).

▼ [Answer](#) ▼ [Solution](#)

Here is a solution that builds on the last example and paves the path toward the general solution presented in the next unit.

$$\begin{aligned} \tilde{\kappa} &= \langle \tilde{\kappa} = [x^T y] \rangle \\ &= \langle \text{each suboperation is performed in floating point arithmetic} \rangle \\ &= \langle \text{reformulate so we can use result from Example 6.3.1.1} \rangle \\ &= \langle \text{use Example 6.3.1.1; twice SCM} \rangle \\ &= \left(\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} + [\chi_2\psi_2] \right) \\ &= \left(\begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) & 0 \\ 0 & (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \end{pmatrix} \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \right. \\ &\quad \left. + \chi_2\psi_2(1 + \epsilon_*^{(2)})(1 + \epsilon_+^{(2)}) \right) \\ &= \langle \text{distribute, commute} \rangle \\ &= \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)}) & 0 \\ 0 & (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)}) \end{pmatrix} (1 + \epsilon_+^{(2)}) \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix} \\ &\quad + \chi_2(1 + \epsilon_*^{(2)})(1 + \epsilon_+^{(2)})\psi_2 \\ &= \langle \text{(perhaps too) slick way of expressing the final result} \rangle \\ &= \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)})(1 + \epsilon_+^{(2)}) & 0 & 0 \\ 0 & (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)})(1 + \epsilon_+^{(2)}) & 0 \\ 0 & 0 & (1 + \epsilon_*^{(2)})(1 + \epsilon_+^{(2)}) \end{pmatrix} \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{pmatrix}, \\ &\text{where } |\epsilon_*^{(0)}|, |\epsilon_*^{(1)}|, |\epsilon_+^{(1)}|, |\epsilon_*^{(2)}|, |\epsilon_+^{(2)}| \leq \epsilon_{\text{mach}}. \end{aligned}$$

$$\begin{aligned} &= \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix}^T \begin{pmatrix} (1 + \epsilon_*^{(0)})(1 + \epsilon_+^{(1)})(1 + \epsilon_+^{(2)}) & 0 & 0 \\ 0 & (1 + \epsilon_*^{(1)})(1 + \epsilon_+^{(1)})(1 + \epsilon_+^{(2)}) & 0 \\ 0 & 0 & (1 + \epsilon_*^{(2)})(1 + \epsilon_+^{(2)}) \end{pmatrix} \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \end{pmatrix} \\ &\text{where } |\epsilon_*^{(0)}|, |\epsilon_*^{(1)}|, |\epsilon_+^{(1)}|, |\epsilon_*^{(2)}|, |\epsilon_+^{(2)}| \leq \epsilon_{\text{mach}}. \end{aligned}$$